

H.W
What are the various problems which we have observed in our society. Pick any few problems, find out they can be solved them by Data Structures and Algorithms.

Ans. 1. Traffic congestion.

- Heavy traffic causes delays and accidents.

- Solution:- use Graphs and shortest path Algorithms (Dijkstra's Algorithm to find the fastest route and reduce Traffic.

2. Hospital Patient Management.

- Problem: Long waiting times and poor patient record management.

Solution:- Use queues to manage patient appointments and trees / Databases to store and retrieve patient records quickly.

3. Waste collection

- Problem:- Garbage is not collected efficiently

- solution: Use Graphs and routes Optimization Algorithm to find the shortest route for garbage collection vehicles

4. Online Shopping Delivery.

- Problem:- Delayed product delivery

- solution: Use Graphs and shortest path Algorithms to select the best delivery route. Use sorting Algorithms to prioritize orders

5. Emergency services (Ambulance / Fire)

- Problem: Emergency vehicles take longer to reach the location

- solution: Use Graphs, GPS and Dijkstra's Algorithm to identify the quickest path and reduce response time.